Table 7.1 System Testing Results

SL No	Description	Input	Expected Output	Actual Output	Status
1	Face Detection Test	Live RTSP stream from Room 408 (18 students)	Faces detected with bounding boxes & 5 landmarks	All 18 faces detected accurately	Pass
2	Blur Rejection Test	Motion-blurred head crop (sigma^2 < 120.0)	Low-quality frames rejected before embedding	184 blurred crops rejected, clean log	Pass
3	Feature Extraction	Aligned 112x112 facial crop	512D unit L2-normalized ArcFace embedding	Vector generated with norm = 1.000	Pass
4	Biometric Matching	Enrolled student probe vector	Cosine sim >= 0.65, identity confirmed	Cosine sim 0.842 >= 0.65, student matched	Pass
5	FAISS Vector Query	Probe vector queried against 10,000 vectors	Top-1 nearest neighbor in < 2.0 ms	Top match returned in 1.1 ms	Pass
6	Candidate Clustering	250 unassigned face appearance vectors	Recurring appearances grouped into clusters	29 clusters formed with >98% purity	Pass
7	Conflict Prevention	Assigning mapped student to new cluster	Old conflicting cluster released, audit logged	Old mapping released, audit record saved	Pass
8	Headcount Normalization	Room 408: 18 attendees vs 59 batch roster	18 marked Present, 41 marked Absent	Exact count: 18 Present / 41 Absent	Pass
9	Spatio-Temporal Check	Same student detected simultaneously in 2 rooms	Dual-presence collision flagged, alert raised	Flagged as impossible, security alert sent	Pass
10	Engagement Analytics	Continuous downward head tilt > 15s with phone pose	Distraction event logged and alert sent	WebSocket toast alert delivered in 180 ms	Pass
11	Role Guard Security	Student account requesting /api/v1/admin/audit-logs	HTTP 403 Forbidden, unauthorized access blocked	HTTP 403 Forbidden returned	Pass
12	End-to-End Latency	Concurrent 18-student live frame processing	Total pipeline latency <= 60.0 ms	Mean latency recorded at 46.5 ms (21.5 FPS)	Pass